

THE RATTY2TAILS PAPERS

MONETARY HISTORY SERIES

The Stone That Cannot Be Quarried

Bitcoin, fixed supply, and the mechanics of price discovery

Every priced object humans have historically traded — gold, oil, equities, real estate, even a fast-food cheeseburger — answers strong demand with more supply. Bitcoin cannot. This essay traces what happens to price when the supply side of the equation is fixed by protocol rather than by production, and asks what four cycles of repricing — 2013, 2017, 2021, and 2025 — actually reveal about how that mechanism behaves in practice.



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TL;DR

- Every asset humans have historically priced — gold, oil, corporate stock, real estate, even a McDonald's cheeseburger — has an elastic supply. Given enough demand, more of it gets made.
- Bitcoin's issuance is fixed by protocol at 21 million coins, released on a known, decreasing schedule enforced by a decentralised network. No government, company, or amount of capital can accelerate it.
- Because new supply cannot be conjured, every marginal buyer has to be satisfied by persuading an existing holder to sell — and the only lever available to do that is price.
- Bitcoin's price history is therefore a series of repricing events: 2013, 2017, 2021 and 2025 each saw the market discover, in a matter of weeks, what price it would take to pry coins out of cold storage.
- Bitcoin's gains are famously concentrated in a handful of trading days a year. Research from analysts such as Econometrics shows that missing the best days of a cycle — not the ordinary volatility in between — is what separates a holder's return from a trader's.
- Inelastic supply is not a one-way ratchet. Bitcoin fell roughly 50 percent from its October 2025 high within about four months, and closed out 2025 as one of its weakest "green" years on record — even as gold quietly had its best year since 1979.

INTRODUCTION & SCOPE

A question every market answers the same way — except one

There is a question that every commodity market answers identically, and Bitcoin answers differently — and the entire shape of Bitcoin's price behaviour follows from that single divergence. The question is this: when more people want a thing than can currently obtain it, where does the extra supply come from? For nearly everything humans have historically priced, the answer is that someone makes more of it. Higher gold prices call new mines into production. Higher oil prices justify new wells. A popular stock can be issued in greater quantity by the company that sells it. A

city block that becomes valuable enough gets rezoned and built taller. This is the ordinary operation of a market economy, and it works quietly, almost invisibly, most of the time.

Bitcoin is the one widely-held monetary asset where that channel does not exist. Its issuance schedule was fixed at launch, is enforced by thousands of independently-operated nodes rather than by any single authority, and cannot be sped up by demand, by capital, or by political will. This essay is a working-through of what that single structural fact implies for price: not as an article of faith, but as an economic mechanism with a traceable history, several genuinely large drawdowns, and at least one uncomfortable caveat that most commentary on the subject prefers to skip. The argument draws on and expands a piece of market commentary the editor came across in researching this series, recasting its central claim — that Bitcoin has no supply-side response to demand — against the historical price record and the growing body of research on where Bitcoin's returns actually come from.

I.

The economics of things that can be made

Start with gold, because it is the asset Bitcoin is most often compared to, and because the comparison is instructive precisely where it breaks down. Gold is scarce in the sense that matters colloquially — there is a finite amount of it in the earth's crust, and no one can manufacture it from nothing. But gold's *annual* supply is not fixed at all. It is a function of price. When the gold price rises, previously uneconomic ore bodies become worth extracting, mining companies raise capital to expand production, and exploration budgets increase. The result, over the twentieth and twenty-first centuries, has been a remarkably steady above-ground stock growth of roughly one to two percent a year — new supply arriving more or less in step with demand, which is a large part of why gold's price has historically been far less volatile than commodities with genuinely inelastic supply.

The same logic runs through every other asset class an investor might reach for as a store of value or a growth vehicle. Enough demand for oil brings new wells online. Enough demand for a company's shares and the company itself can print more of them — a mechanism some Bitcoin-holding public companies have used quite deliberately in recent years, issuing equity specifically to buy more Bitcoin with the proceeds, which is itself a small illustration of the asymmetry this essay is about. Enough demand for burgers and the kitchen fries another batch. Enough demand for

a particular penthouse and, eventually, someone finds a way to build another floor. In every one of these cases, the market has two variables to adjust — price *and* quantity — and it typically uses both, which is exactly what keeps any single price move from becoming too extreme for too long.

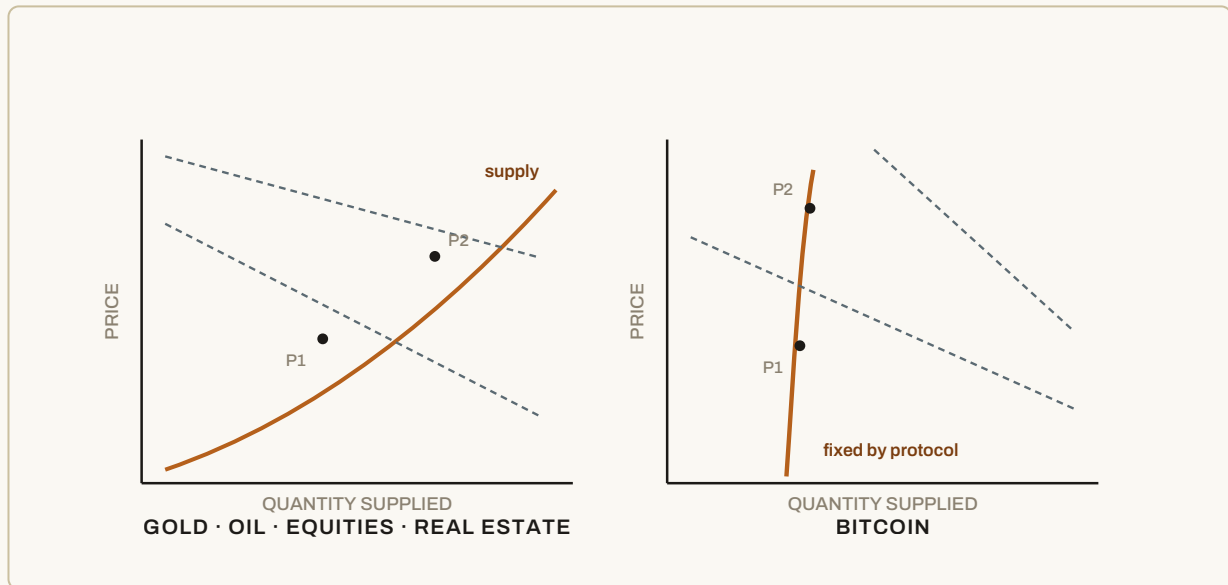


Fig. 1. Two supply curves. Left: an ordinary asset, where a demand shock ($D1 \rightarrow D2$) is absorbed partly by higher price and partly by new production, so quantity supplied rises alongside price. Right: an asset with a supply fixed by protocol, where the identical demand shock cannot call forth new quantity at all — the entire adjustment has to show up in price.

This is the textbook mechanism, and it is worth stating plainly because it is easy to take for granted: **price is normally not the only thing that moves when demand changes. Quantity moves too.** That second channel is what Bitcoin does not have.

II.

A supply schedule fixed in code

Bitcoin's issuance is set by its protocol, not by any producer's judgement about whether current prices justify more output. New coins are created only as a reward paid to miners for extending the blockchain, at a rate that halves roughly every four years — a schedule fixed in 2009 and adhered to, without exception, by every version of the software that the network's participants have ever run. The reward began at 50 BTC per block; by 2024 it had halved four times, to 3.125 BTC; the total supply

approaches, but by design never quite reaches, 21 million coins, with the overwhelming majority already in circulation and the remainder trickling out on a schedule that runs into the twenty-second century.

No central bank votes on this. No mining company's decision to expand or idle its operations changes it — more computing power thrown at the network shifts *which* miner earns each block reward, and adjusts the network's mining difficulty to keep block times roughly constant, but it does not change *how many* new coins are issued per block. A government could not print more of it into existence even if it wished to, short of persuading the majority of the network's economically-relevant participants to adopt a rule change that increases the supply cap — something that would require an extraordinary, coordinated departure from the property that gives the asset its scarcity in the first place, and which the network's history of rejecting supply-increasing proposals suggests is extremely unlikely.

The practical consequence is that the classic supply-and-demand diagram has to be redrawn for Bitcoin. Where gold's supply curve slopes gently upward with price, Bitcoin's supply curve — at any given moment, for the coins already mined — is close to vertical. Not perfectly vertical: the schedule of new issuance still adds a small, known, ever-shrinking trickle each day. But for any demand shock large enough to matter, that trickle is negligible against the size of the move. The overwhelming majority of the "supply" that meets a new wave of buyers has to come from somewhere else entirely: from an existing holder — someone who already owns coins mined years ago — deciding that the current price is finally attractive enough to sell.

If demand cannot be met by making more of the thing, it can only be met by paying an existing owner enough to change their mind.

— THE MECHANISM, RESTATED PLAINLY

III.

Price as the only adjustable variable

This reframes what a Bitcoin order book actually represents. At any given price, some quantity of existing holders is willing to sell — coins sitting in exchange wallets, or in cold storage but earmarked for sale above some threshold the owner has in mind. If buy pressure exceeds that quantity, the price simply has to rise until it reaches a level where enough additional holders decide they, too, are willing to part with their coins.

There is no factory floor to speed up, no additional rig to spin up, no zoning approval to seek. The market clears entirely on the price axis. Economists sometimes call the resulting process *price discovery*: not the market converging on some pre-existing "correct" value, but the market actively searching, in real time, for the price at which enough sellers appear to satisfy the buyers who have shown up.

Bitcoin's trading history reads, in this light, as a series of discrete price-discovery episodes rather than a smooth trend. Four stand out. In 2013, the price ran from roughly one hundred dollars to a peak near \$1,150 before collapsing to around \$200 amid the collapse of the Mt. Gox exchange. In 2017, it ran from under \$1,000 to nearly \$19,700 by December, before a year-long decline to roughly \$3,200. The 2020–21 cycle carried it from pandemic-era lows to an all-time high near \$69,000 in November 2021, followed by a fall to around \$15,500 by late 2022. Most recently, a rally through 2025 — driven substantially by sustained inflows into newly-launched U.S. spot Bitcoin exchange-traded funds — pushed the price to a fresh all-time high just above \$126,000 on 6 October 2025, before a correction that had cut the price roughly in half, to around \$60,000, by February 2026, and which has left it trading in the low-to-mid \$60,000s through the summer of 2026.

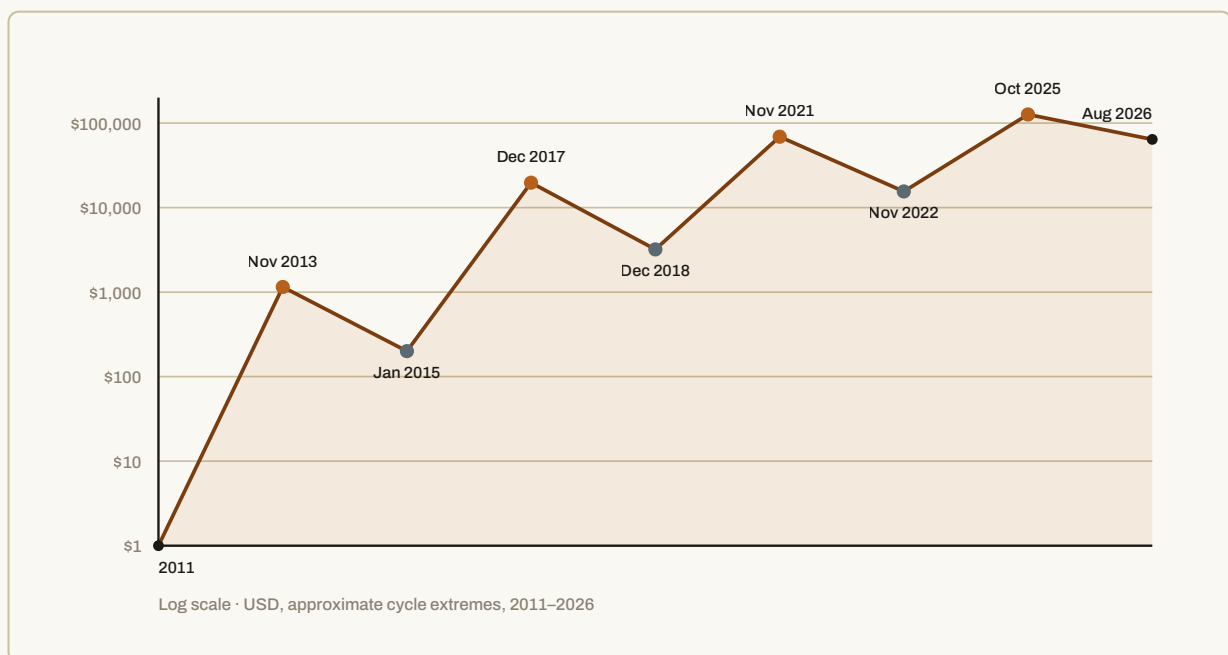


Fig. 2. Bitcoin, approximate cycle extremes, 2011–2026, log scale. Each cycle's peak and trough is separated by an order of magnitude or more — the signature, this essay argues, of a market that can only clear through price, run in both directions.

What makes this pattern notable is not merely that the price has risen a great deal over fifteen years — plenty of speculative assets have done that and then stayed up, or drifted down slowly. It is the shape of each individual move: rapid, front-loaded appreciation as new demand searches for a market-clearing price against a supply that will not budge, followed by an equally rapid unwind once a critical mass of recent buyers — the ones with the least conviction and the shortest time horizon — decide to become sellers themselves. Both halves of that pattern are consequences of the same underlying fact. A market that cannot manufacture new supply to meet demand also cannot manufacture new supply to absorb a wave of selling. Inelastic supply cuts in both directions.

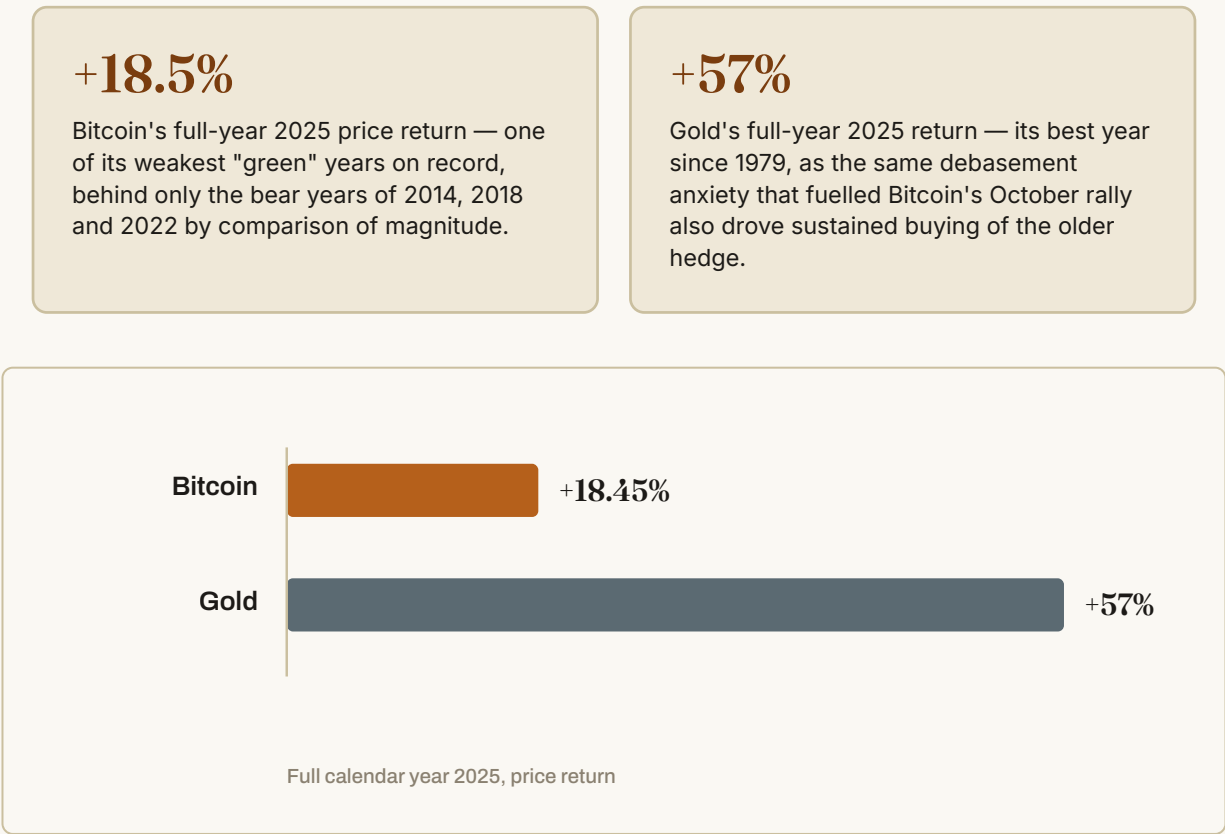
IV.

Debasement, liquidity, and an uncomfortable footnote from 2025

The demand side of this story is not accidental, and it is worth being explicit about where it tends to come from. A meaningful share of the buying that has driven each of Bitcoin's major rallies has coincided with periods of expansionary monetary policy or growing anxiety about government deficits and debt monetisation — the kind of environment in which holders of cash-like assets go looking for something that a central bank or treasury cannot simply create more of. This is a very old idea in monetary economics, sometimes discussed under the heading of the Cantillon effect: newly-created money does not enter the economy evenly, and those positioned to acquire hard assets before the broader price level adjusts tend to benefit at the expense of those holding the depreciating currency itself. Bitcoin's appeal to a segment of investors is precisely that it offers a modern, liquid, bearer-asset version of that old hedge — a way of opting out of currency debasement without the storage and verification costs of physical gold.

2025 supplied a clean illustration of the demand-side mechanics. Concern over U.S. fiscal sustainability, alongside the March 2025 executive order establishing a U.S. Strategic Bitcoin Reserve funded initially from existing forfeiture holdings, coincided with sustained inflows into spot Bitcoin ETFs that had only become available to U.S. investors the previous year. Institutional holdings of Bitcoin climbed toward the high hundreds of billions of dollars over the course of the year, and the price cleared \$120,000 for the first time in July, then \$124,000 in August, then its eventual peak above \$126,000 in early October.

Here the honest version of this essay has to include a detail that undercuts the tidier version of the narrative: 2025, despite that headline all-time high, was actually one of Bitcoin's *weakest* calendar years on a full-year return basis, and it was comprehensively outperformed by the very asset it is often framed as displacing.



V.

The days that matter

A frequently-cited feature of Bitcoin's price behaviour, and one directly related to the mechanics described above, is that its returns are heavily concentrated in a small number of trading days. Because price moves in sharp, front-loaded bursts as the market searches for a new equilibrium, a disproportionate share of any cycle's total gain tends to occur within a handful of sessions — commentary from analysts including Bitwise's Hunter Horsley and research published by outlets such as PlanC and Econometrics has repeatedly found that somewhere in the region of ten trading days a year account for the bulk of Bitcoin's annual performance.

Econometrics' analysis of the first full cycle following the 2012 halving illustrates just how large the effect can be. A holder who stayed invested throughout that cycle saw roughly a 49-fold increase in value from the halving to the cycle's peak. A holder who missed only the best five days of that same cycle saw the multiple fall to roughly 13-fold — a reduction of about three-quarters. A holder who missed the best ten days saw it fall further still, to roughly five-fold — a reduction of nearly ninety percent, from missing a period equivalent to a couple of long weekends spread across four years.

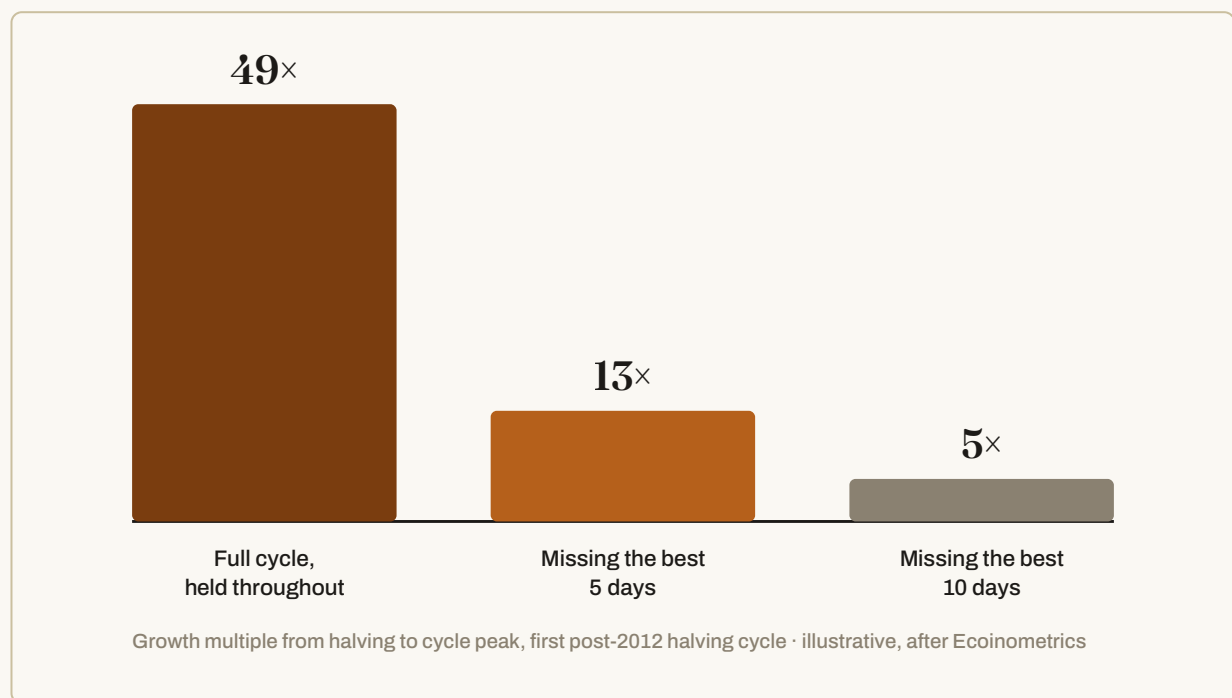


Fig. 3. The cost of missing the best days of a cycle, first post-2012-halving cycle. Multiples are illustrative of the order of magnitude Econometrics reported and will vary somewhat by cycle and methodology; the direction and scale of the effect are the point.

This concentration effect follows directly from the supply mechanics discussed earlier. If price only moves when a critical threshold of demand overwhelms available supply, then most of a cycle's gain should indeed be compressed into the specific windows where that threshold is crossed — a large ETF inflow day, a surprise policy announcement, a moment when a wave of buyers arrives faster than existing holders are willing to sell into it — rather than being spread evenly across the calendar. The practical upshot, repeated by nearly every serious long-term Bitcoin investor in one form or another, is that trying to time entries and exits around this asset is a considerably higher-stakes bet than it is in most other markets, precisely because the distribution of returns is so lopsided.

VI.

Strong hands and the cost of admission

There is a behavioural corollary to all of this that is worth drawing out explicitly. Because the only variable available to clear the market is price, and because price moves in concentrated, unpredictable bursts, the asset simply cannot offer a smooth ride to whoever holds it. On-chain analysis of coin age — tracking how long Bitcoin sitting in a given wallet has gone unmoved — consistently shows the same pattern across every cycle: the share of the supply held by long-term, low-turnover owners rises through bear markets, as short-term speculators are shaken out and the coins that remain concentrate among holders with a multi-year time horizon, and then falls again during the blow-off phase of a bull market, as some of those long-term holders finally sell into the strength that price discovery has produced. Each cycle effectively redistributes coins from participants with a low pain threshold to participants with a higher one — which is one reason Bitcoin advocates so often describe volatility not as a defect to be engineered away, but as the actual mechanism by which the asset's ownership base has become progressively harder to dislodge.

That framing is honest as far as it goes, but it should not be mistaken for a promise that volatility is costless. A near-vertical supply curve means the market has no shock absorber in either direction. There is no equivalent, on the way down, of an oil producer shutting in wells until prices recover, or a gold miner idling a high-cost pit — no mechanism by which falling prices reduce the flow of coins onto the market and thereby cushion the fall. Sellers sell into whatever bids exist, exactly as buyers buy from whatever offers exist on the way up. The same structural feature that produces a 49-fold cycle also produces a 50 percent drawdown in four months. An investor drawn

to Bitcoin by the fixed-supply argument should understand that they are choosing into both halves of that trade, not just the favourable one — and that the "cost of admission," as some in the space describe the drawdowns, is not a bug that a future, more mature version of the asset is likely to eliminate. It follows directly from the same mechanism that gives the asset its scarcity in the first place.

SUMMARY

What the fixed supply actually buys you

Bitcoin's central economic distinction is not that it is scarce — gold is scarce too, and so, in the relevant sense, is a first-edition book or a piece of beachfront land. It is that Bitcoin's scarcity is fixed independently of price, enforced by a decentralised network rather than by the economics of extraction, and therefore incapable of the supply-side response that moderates price movements in almost every other market. That single structural difference explains why Bitcoin's history looks like a sequence of sharp repricing events rather than a gentle trend; why its returns are so concentrated in a small number of trading days each cycle; and why the same mechanism that produces spectacular multi-year gains also produces drawdowns of fifty percent or more with no shock absorber built in.

None of this constitutes investment advice, and none of it resolves the genuinely open question of how much demand for a debasement hedge will continue to show up in any given year — 2025's gap between Bitcoin's middling calendar-year return and gold's best year in four and a half decades is a useful reminder that fixed supply guarantees a mechanism, not an outcome. What it does offer is a more precise vocabulary for a question that gets asked, usually informally, every time the price moves sharply in either direction: not "why is Bitcoin so volatile," but "why would we ever expect an asset with this supply schedule to be anything else."

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This essay is published as part of the Monetary History Series within The Ratty2Tails Papers, a long-form imprint covering monetary economics and history through to Bitcoin. It was adapted and expanded by the editor from market commentary reviewed during research for this series, cross-checked against the historical price record and cited third-party analysis. Figures are stated as of August 2026 and are approximate where sourced from contemporaneous market reporting.